

Zach Sweeney → Exam 1

Problem 1

78

Given $G = 0.19$ $P = 14.3 \text{ g/cc}$ $\sigma_p = 587 \cdot 10^{-24} \text{ cm}^2$

Find Q $\phi = 4 \cdot 10^{12}$

U-N fuel

$$N_{235} = \frac{P N_A G}{M_{UN}}$$

$$Q = E_f \cdot \underbrace{N_p \cdot \sigma_p \cdot \phi}_{N_{UN}} \quad M_{UN} = (0.19 \cdot 235 + 0.81 \cdot 238) + 14$$

$$M_{UN} = 251.43$$

$$N_{235} = \frac{14.3 \text{ g/cc} \cdot 6.022 \cdot 10^{23} \frac{\text{atoms}}{\text{mol}} \cdot \frac{1 \text{ atom U}}{1 \text{ molecule UN}} \cdot 0.19}{251.43}$$

$$N_{235} = 6.51 \cdot 10^{21} \frac{\text{U235}}{\text{cm}^3}$$

$$\therefore Q = E_f \cdot N_p \cdot \sigma_p \cdot \phi$$

$$Q = 200 \cdot 10^6 \text{ eV} \cdot 1.6 \cdot 10^{-19} \frac{\text{kJ}}{\text{eV}} \cdot 6.51 \cdot 10^{21} \cdot 587 \cdot 10^{-24} \cdot (4 \cdot 10^{12} \frac{1}{\text{cm}^2 \cdot \text{s}})$$

$$Q = 4789 \frac{\text{W}}{\text{cc}}$$

Part 2 - For UO_2 , $P = 10.97$

$$N_{235} = \frac{P N_A G}{M} = \frac{10.97 \cdot 6.022 \cdot 10^{23} \cdot X}{M_{\text{UO}_2}}$$

$$M_{\text{UO}_2} = (X \cdot 235) + (1-X) \cdot 238 + 2 \cdot 16$$

$$N_{235} = \frac{10.97 \cdot 6.022 \cdot 10^{23} \cdot X}{235X + 238 - 238X + 32}$$

Problem 1 (cat)

2

Simplifying $N_{235} \rightarrow N_{235} =$

$$\frac{6.61 \cdot 10^{24} \cdot X}{-3X + 270}$$

where $X = \text{enrichment}$

Now, plugging into heat eqn:

$$Q = E_F N_F \sigma_F \phi \rightarrow N_F = \frac{Q}{E_F \phi \sigma_F}$$

$$\frac{6.61 \cdot 10^{24} X}{-3X} + \frac{X}{270} = \frac{Q}{E_F \phi \sigma_F}$$

$$X = 270 \cdot \left(\frac{Q}{E_F \phi \sigma_F \cdot 2.203 \cdot 10^{21}} \right)$$

Plugging in $\rightarrow$

$$X = 270 \cdot \frac{489 \frac{\text{W}}{\text{cc}}}{200 \cdot 10^6 \cdot 1.6 \cdot 10^{-14} \cdot 587.5^{21} \cdot 4 \cdot 10^{12}}$$

$X = 0.176$

 $\rightarrow$ this seems too high

- you lost me somewhere

- should be higher...

- could set $N^{235}(\text{U}) = N^{235}(\text{UO}_2)$

Problem 2

3

Given $u_c = 0.18$ $u_p = 0.01$ $h_{cool} = 1.5$

$Q = 350$ $R_f = 0.45$ $t_g = 80 \mu m$ $t_c = 0.06 cm$

$T_{cool} = 540 K$

T_{cool} is known, work outside-in

$$T_{oc} - T_{cool} = \frac{Q}{2h_{cool}} R_f = \frac{LHR}{2\pi R_f h_{cool}}$$

First, calculate LHR $\rightarrow LHR = \pi R^2 Q$

$$LHR = \pi (0.45)^2 350$$

$$LHR = 222 \frac{W}{cm^2}$$

$$T_{oc} = T_{cool} + \frac{LHR}{2\pi R_f h_{cool}} = \frac{222}{2\pi \cdot 0.45 \cdot 1.5} + 540 K$$

$$T_{oc} = 592 K$$

$T_{oc} \rightarrow T_{midpt} \rightarrow$

$$T_{ic} - T_{oc} = \frac{\cancel{LHR}}{\cancel{2\pi R_f h_{cool}}} \frac{LHR}{2\pi R_f} \frac{t_g}{u_c}$$

here $t_c = t_g - r$

$$T_{0.1} = T_{oc} + \frac{LHR}{2\pi R_f} \cdot \frac{t_{0.1}}{u_c}$$

midpoint of cladding

$$T_{0.1} = 592 K + \frac{222 \frac{W}{cm^2}}{2\pi \cdot 0.45} \cdot \frac{0.03}{0.18}$$

$$T = 697 K$$

= calculator error?

- looks right,
but wrong value...

$$T @ r = 0.1 ?$$

Problem 3 $L = 3.6 \text{ m}$

Given $LHR^0 = 400 \frac{\text{W}}{\text{cm}}$ $\gamma = 1.25$

$C_p = 4200 \frac{\text{J}}{\text{kgK}}$ $\dot{m} = 0.1 \frac{\text{kg}}{\text{s}}$ $T_i = 580 \text{ K}$

Find LHR @ $z = 2.1 \text{ m}$ (Axial ~~heat~~ error)

$z_0 = \frac{L}{2} = \frac{3.6}{2} = 1.8 \text{ m}$

$LHR \frac{z}{z_0} = LHR^0 \cos \left[\frac{\pi}{2\gamma} \left(\frac{z}{z_0} - 1 \right) \right]$

$LHR(2.1) = 400 \cdot \cos \left[\frac{\pi}{2(1.25)} \left(\frac{2.1}{1.8} - 1 \right) \right]$

$LHR(2.1) = 391 \frac{\text{W}}{\text{cm}}$

Now Find T_{cool}

$T_{cool} - T_{cool}^i = \frac{2\gamma}{\pi} \cdot \frac{z_0 LHR^0}{\dot{m} C_p} \left\{ \sin \frac{\pi}{2\gamma} + \sin \left(\frac{\pi}{2\gamma} \left(\frac{z}{z_0} - 1 \right) \right) \right\}$

at outlet, $z = 3.6$

$T_{cool}^{out} = T_{cool}^i + \frac{2\gamma}{\pi} \cdot \frac{z_0 LHR^0}{\dot{m} C_p} \left\{ \sin \left(\frac{\pi}{2\gamma} \right) + \sin \left(\frac{\pi}{2\gamma} \left(\frac{z}{z_0} - 1 \right) \right) \right\}$

Plug in $\rightarrow$

$T_{cool}^{out} = 580 \text{ K} + \frac{2 \cdot 1.25}{\pi} \cdot \frac{1.8 \cdot 400}{0.1 \cdot 4200} \left\{ \sin(1.25) + \sin \left(\frac{\pi}{2 \cdot 1.25} \left(\frac{3.6}{1.8} - 1 \right) \right) \right\}$

$T_{out} = 580 \text{ K} + 25.9 \text{ K} = 606 \text{ K}$ $\rightarrow$ coolant outlet

Seems like a very small ΔT , could be units error

Problem 4Given 5% bump FIMA, $T_f = 1200\text{ K}$

w/ FIMA

$$K_{\text{on}} = \frac{1}{A + BT}$$

$$A = 3.8 + 200 \cdot 0.05 = 13.8$$

$$B = 0.0217$$

$$K_{\text{on}} = \frac{1}{13.8 + 0.0217 \cdot 1200\text{ K}}$$

$$K_{\text{on}} \text{ w/ FIMA} = 0.0251 \frac{\text{W}}{\text{cm}^2\text{K}}$$

w/o FIMA $\rightarrow$ ~~K_{on}~~

$$A = 3.8 + 200 \cdot \overset{\rightarrow 0}{\text{FIMA}} = 3.8$$

$$K_{\text{on}} = \frac{1}{3.8 + 0.0217 \cdot 1200}$$

$$K_{\text{on}} = 0.0335 \frac{\text{W}}{\text{cm}^2\text{K}}$$

(No FIMA)

 $\therefore$ FIMA accounts for about a ~~40%~~ % reduction $\hookrightarrow 25-35\%$

5/6 5) Steady-state, Axisymmetric, Constant T ,
and k is independent of T

6

5/5 6) UF_6 is used for enrichment. For centrifuge, gas is flown into rotating vessel, where enriched U-235 flows to the center and is extracted. Depleted U is heavier, moves to outside.

6/8 7.) DNBR is Ratio of critical heat flux to Actual heat flux. Critical heat flux is when flow is in transition boiling. ~~FNBR~~ This needed more...

2/4 8.) ΔT_{max} density is a ratio of fuel volume to cladding internal volume. This ensures proper heat flow, and that cladding or gap are sufficient size to allow proper flow. - nope...

1/4 a.) Swelling, Creep
- not really...

4/6 10.) Cladding $\rightarrow$ contains fuel pellets in uniform space
 $\hookrightarrow$ contains fission products. ^{somewhat...}

Zr is effective since it has a low neutron capture cross-section. - more here...

4/4 11.) Fuel Rod, cladding, gap, ~~some~~ coolant
 $\hookrightarrow$ technically, more of a secondary role

- 12.) 1. how effective fuel transfers heat to coolant ✓
2. behavior during Normal operations ✓
3. behavior during accidents / Accident tolerance ✓

13.) Finite Element → Any geometry / BC ✓
→ Complicated ✓

Finite Volume → Can model any geometry ✓
→ Limited on BC's ✓

Finite Difference → simplest ✓
→ can only handle basic geometry ✓

14.) ⊕ Very Inexpensive
Years of OE, predictable

- needed specifics here

⊖ Brittle ✓
Not the best thermal conductor ✓

15.) ~~Implicit~~ → solves using present data point

~~Explicit~~ → solves
→ backward Euler

Implicit → future data point used ✓

explicit → Forward Euler
→ current state used to start ✓

- wanted a little more